ITEM-f: a self-contained convergence proof via performance estimation

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1 Introduction

ITEM-f is a fixed-step first-order method for minimizing an L-smooth and μ-strongly convex function $f\colon \mathbb{R}^d \to \mathbb{R}$. After N gradient evaluations it returns a point x_N with

$$f(x_N) - f_\star \leq \frac{1}{\Upsilon_N^2}\big(f(x_0) - f_\star\big) \leq 4(1-\sqrt{q})^{2N}\big(f(x_0) - f_\star\big),$$

where $x_\star$ is the minimizer of f, $q = \mu/L$, and the constant Υ_N is defined by a planar geometric construction. This document is a self-contained supplement to the companion paper [1], where the method is introduced: nothing here depends on the companion paper, and the companion paper also contains a Lyapunov-function proof of the same theorem, which is not reproduced here.

The proof strategy is the performance estimation (PEP) methodology [3, 6]. We formulate the worst-case performance of the method as an optimization problem, relax it to a convex semidefinite program (SDP), pass to the dual SDP, and exhibit an explicit analytic feasible point of that dual whose objective value is $1/\Upsilon_N^2$. Weak duality converts the feasible point into the convergence guarantee; strong duality is never invoked. Because the certificate is built for a stepsize-matrix representation of the method, the final section proves that this representation coincides with the momentum form in which ITEM-f is stated.

The document is organized as follows. Section 2 fixes notation. Section 3 states the algorithm, the geometric construction that supplies its coefficients, and the convergence theorem. Section 4 proves that the construction exists, is unique, is computable by bisection, and satisfies the identities the proof consumes. Section 5 develops the PEP formulation and its dual. Section 6 states the analytic certificate, Section 7 verifies its feasibility, and Section 8 closes the argument by proving the equivalence of the certified method to ITEM-f. Section 9 records how the certificate was found.

2 Notation

Write $\mathbb{R}^d$ for the underlying Euclidean space and write $\langle\cdot,\cdot\rangle$ and $\|\cdot\|$ for the standard inner product and norm on $\mathbb{R}^d$; for planar vectors $u, v \in \mathbb{R}^2$ we also write $u \cdot v$ for the inner product and $\det(u,v) \triangleq u_1v_2 - u_2v_1$. Write $\mathbb{R}^{m\times n}$ for the set of $m \times n$ matrices, $\mathbb{S}^n$ for the set of $n\times n$ symmetric matrices, and $\mathbb{S}^n_+$ for the set of $n \times n$ positive-semidefinite matrices. We use the standard notation e_i for the unit vector having a single 1 as its i-th component. Write $(\cdot \odot \cdot) : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{S}^n$ to denote the symmetric outer product, that is, for any $x, y \in \mathbb{R}^n$, $x \odot y = (xy^\intercal + yx^\intercal)/2$, and $\delta_{i,j}$ denotes the Kronecker delta. We write $\mathbf{tr}$ for the trace of a square matrix.

For $0 \leq \mu < L < \infty$, we write $\mathcal{F}_{\mu,L}$ for the class of L-smooth μ-strongly convex functions $f\colon \mathbb{R}^d \to \mathbb{R}$, that is, differentiable functions for which ∇f is L-Lipschitz and $f - (\mu/2)\left\|\cdot\right\|^2$ is convex; the case $\mu = 0$ gives the class $\mathcal{F}_{0,L}$ of L-smooth convex functions. Throughout, $x_\star$ denotes the minimizer of the function f under consideration, $f_\star \triangleq f(x_\star)$, and $q \triangleq \mu/L \in (0,1)$ is the inverse condition number. For a method that runs for N iterations and queries the points $x_0, x_1, \ldots, x_N$, we collect the iteration indices together with the optimal point in the index set $I_N^\star \triangleq \{0, 1, \ldots, N, \star\}$. Finally, we use the single-gradient-step notation $x^+ \triangleq x - \frac{1}{L}\nabla f(x)$.

3 The algorithm and its guarantee

Consider the unconstrained minimization problem

$$\underset{x\in\mathbb{R}^d}{\text{minimize}} \quad f(x), \tag{$\mathcal{P}'$}$$

where $f : \mathbb{R}^d \to \mathbb{R}$ is an L-smooth and μ-strongly convex function with $0 < \mu < L$. Strong convexity guarantees that f has a unique minimizer $x_\star$, and denote $f_\star \triangleq f(x_\star)$.

Algorithm statement. The method ITEM-f is

$$x_{k+1} = x_k^+ + \frac{a_k \phi_{N-k-1}}{(1-q)a_{k+1}\phi_{N-k}}(x_k^+ - x_{k-1}^+) + \frac{\phi_{N-k-1}}{\phi_{N-k}}(x_k^+ - x_k), \tag{ITEM-f}$$

for $k = 0, 1, 2, \ldots, N-1$, where N is a prespecified iteration count, $x_0 \in \mathbb{R}^d$ is the starting point, $x_{-1}^+ = x_0$ by convention, $x_k^+ = x_k - \frac{1}{L}\nabla f(x_k)$,

$$q = \mu/L, \quad \phi_0 = \sqrt{1-q}, \quad \phi_N = \frac{(1-q)\Upsilon_N + a_N}{\sqrt{1-q}}, \quad \phi_j = \sqrt{2\Upsilon_N a_j - q}, \quad 1 \leq j \leq N-1,$$

and $a_0, a_1, \ldots, a_N$ and Υ_N are positive constants that we define in Lemma 1.

Taylor and Drori [5, Appendix E] reported numerically optimized step sizes of ITEM-f for $N = 1, \ldots, 5$ in the setting $L = 1$ and $\mu = 0.1$, but did not assign a name to the resulting method. They also derived a closed-form method with a convergence analysis, called the Information-Theoretic Exact Method (ITEM), for a related performance criterion. We use the name *ITEM-f* to acknowledge their numerical construction while distinguishing it from ITEM.

Construction of the coefficients. To fully define ITEM-f, it remains to specify the coefficients $a_0, a_1, \ldots, a_N$ and Υ_N. Lemma 1 formally defines these coefficients, and Figure 1 illustrates the underlying geometric construction.

Lemma 1. *For any $N \geq 1$ and $0 < q < 1$, there exists unique $\Upsilon_N > 1$ and*

$$1/\Upsilon_N = a_0 < a_1 < \cdots < a_N < a_{N+1} = \Upsilon_N$$

satisfying the following conditions. Write $C = (\Upsilon_N, 0)$ and $R_N = \sqrt{\Upsilon_N^2 - 1}$. Let $b_0, b_1, \ldots, b_N, b_{N+1}$ be a positive sequence, write $P_k = (a_k, b_k) \in \mathbb{R}^2$ for $0 \leq k \leq N+1$. The requirements are:

$$\begin{aligned}
&\|P_k - C\|^2 = R_N^2, \quad 1 \leq k \leq N, \\
&(P_k - C)\cdot(P_{k+1} - C) = R_N^2\Big(1 - \tfrac{q\, a_{k+1}}{\Upsilon_N}\Big), \quad 0 \leq k \leq N, \\
&P_0 = \Big(\tfrac{1}{\Upsilon_N}, \tfrac{\sqrt{1-q}\, R_N}{\Upsilon_N}\Big), \\
&P_{N+1} = \big(\Upsilon_N, \sqrt{1-q}\, R_N\big).
\end{aligned} \tag{1}$$

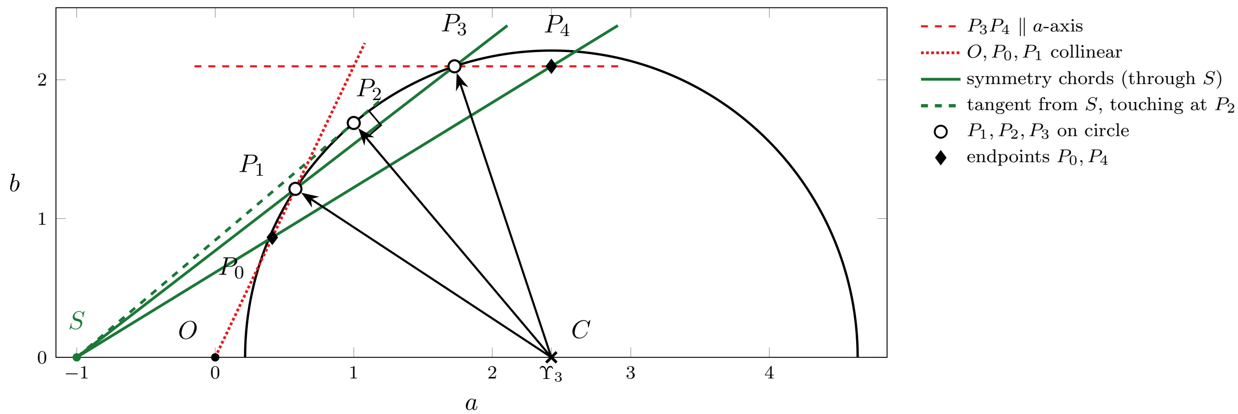

Figure 1: The planar construction underlying ITEM-f, shown for $N = 3$, $q = 0.1$. The interior points $P_1, \ldots, P_N$ ($\circ$) lie on the circle of radius $R_N = \sqrt{\Upsilon_N^2 - 1}$ centered at $C = (\Upsilon_N, 0)$; the endpoints P_0 and P_{N+1} ($\diamond$) lie strictly inside it. The relation $(P_k - C) \cdot (P_{k+1} - C) = R_N^2(1 - q\, a_{k+1}/\Upsilon_N)$ determines each point on the circle. *Backward shooting (red).* The recurrence is run backward from the fixed terminal angle $\theta_N = \arccos(-\sqrt{q})$, i.e. from the horizontal final edge at height $b_N = b_{N+1} = \sqrt{1-q}\, R_N$ (dashed); Υ_N is the value for which O, P_0, P_1 are collinear in this order (dotted). *Symmetry (dark green).* Under $T(P_k) = P_{N+1-k}$ every symmetric pair P_kP_{N+1-k} is collinear with $S = (-1, 0)$; the fixed point P_2 is the contact point of the tangent from S, so $SP_2 \perp (P_2 - C)$.

The proof of Lemma 1 is given in Section 4. To clarify, a_{N+1} is not used in the definition of ITEM-f but we define the notion for convenience. We can compute the numerical values of $a_0, \ldots, a_N$ and Υ_N by a one-dimensional bisection, as discussed in Section 4.

The geometric construction carries a symmetry: the map $T(a, b) \triangleq (1/a,\, b/a)$ maps the circle $a^2 + b^2 - 2\Upsilon_N a + 1 = 0$ to itself and reverses the construction, $T(P_k) = P_{N+1-k}$ for $0 \le k \le N+1$, so that $a_k a_{N+1-k} = 1$ and $a_k b_{N+1-k} = b_k$ (Lemma 6 in Section 4); in Figure 1 this symmetry is visible as the chords P_kP_{N+1-k} passing through the point $S \triangleq (-1, 0)$.

Convergence theorem. The main result of this document is the following guarantee.

Theorem 1. *Let $f : \mathbb{R}^d \to \mathbb{R}$ be L-smooth and μ-strongly convex with $q = \mu/L \in (0, 1)$, and let $x_\star = \operatorname{argmin} f$. Then the final iterate x_N of* (ITEM-f) *satisfies*

$$f(x_N) - f_\star \;\le\; \frac{1}{\Upsilon_N^2}\big(f(x_0) - f_\star\big) \;\le\; 4(1 - \sqrt{q})^{2N}\big(f(x_0) - f_\star\big). \tag{2}$$

The proof of Theorem 1 occupies Sections 5–8, with the second inequality supplied by Lemma 7.

4 The construction: existence, uniqueness, computation

We first derive the target angle and describe the shooting computation of the ITEM-f coefficients, then prove that it produces the unique construction of Lemma 1. Throughout, $N \ge 1$ and $q \in (0, 1)$ are fixed, $\Upsilon > 1$ denotes a candidate value of the constant, and we write

$$C(\Upsilon) \triangleq (\Upsilon, 0), \quad R(\Upsilon) \triangleq \sqrt{\Upsilon^2 - 1}, \quad p(\Upsilon) \triangleq \frac{q\, R(\Upsilon)}{\Upsilon} = q\sqrt{1 - \Upsilon^{-2}},$$

abbreviated to C, R, and p when the argument is clear. The map $\Upsilon \mapsto 1-\Upsilon^{-2}$ is continuous and strictly increasing on $(1,\infty)$; hence so is p, with $0<p(\Upsilon)<q$, $p(\Upsilon)\to 0$ as $\Upsilon\downarrow 1$, and $p(\Upsilon)\to q$ as $\Upsilon\to\infty$.

Target angle. Fix a candidate $\Upsilon>1$. Write $P_1-C=R(\cos\theta_1,\sin\theta_1)$ with $\theta_1\in(0,\pi)$. The endpoint formula gives $P_0-C=\left(-R^2/\Upsilon,\ \sqrt{1-q}\,R/\Upsilon\right)$, so the inner-product condition at $k=0$ evaluates on both sides; equating the two evaluations and simplifying (tedious but straightforward algebra) gives

$$\sin\theta_1=\sqrt{1-q}\,(\Upsilon+R\cos\theta_1)=\sqrt{1-q}\,a_1. \tag{3}$$

Since $b_1=R\sin\theta_1$, this says that P_1 lies on the line $b=\sqrt{1-q}\,R\,a$ through the origin and P_0. Substitution into $(a-\Upsilon)^2+b^2=R^2$, using $\Upsilon^2-R^2=1$, gives the quadratic $\left(1+(1-q)R^2\right)a^2-2\Upsilon a+1=0$, whose roots are $a_\pm=\left(\Upsilon\pm\sqrt{q}\,R\right)/\left(1+(1-q)R^2\right)$. The quadratic is negative at $a_0=1/\Upsilon$, so $a_-<a_0<a_+$. Hence the ordering constraint selects the point $(a_+,\sqrt{1-q}\,R(\Upsilon)a_+)$. Its angle about $C(\Upsilon)$ is the *target angle* $\vartheta(\Upsilon)\in(0,\pi)$, where

$$\cos\vartheta(\Upsilon)\triangleq\frac{a_+-\Upsilon}{R}=\frac{\sqrt{q}-(1-q)R\,\Upsilon}{1+(1-q)R^2}. \tag{4}$$

We also define $\vartheta(\Upsilon)$ for any $\Upsilon>1$ by this formula, even if the corresponding construction does not exist.

Shooting computation. For each candidate $\Upsilon>1$, define its backward shooting orbit by

$$\begin{aligned}\theta_N(\Upsilon)&\triangleq\arccos(-\sqrt{q}),\\ \theta_k(\Upsilon)&\triangleq\theta_{k+1}(\Upsilon)+\arccos(1-q-p(\Upsilon)\cos\theta_{k+1}(\Upsilon)),\quad k=N-1,\dots,1.\end{aligned} \tag{5}$$

Every admissible construction obeys this recurrence. Indeed, if a candidate Υ admits the required points, then every interior point has a unique representation $P_k-C(\Upsilon)=R(\Upsilon)(\cos\theta_k,\sin\theta_k)$ with $\theta_k\in(0,\pi)$. The ordering of the abscissas gives $\theta_1>\cdots>\theta_N$, while the inner-product conditions give

$$\cos(\theta_k-\theta_{k+1})=1-q-p(\Upsilon)\cos\theta_{k+1}$$

for $1\le k\le N-1$. At $k=N$, the same condition gives $\sin\theta_N=\sqrt{1-q}$, and $a_N<a_{N+1}$ selects $\cos\theta_N=-\sqrt{q}$. Therefore admissible constructions satisfy (5).

Define the shooting residual by

$$\Xi_N(\Upsilon)\triangleq\theta_1(\Upsilon)-\vartheta(\Upsilon)$$

for $\Upsilon>1$. Lemma 5 below shows that the candidate closes exactly when $\Xi_N(\Upsilon)=0$.

The numerical value of the root Υ_N can be computed by bisection. Start with $\Upsilon_l=1$ and $\Upsilon_r=2$, double Υ_r until $\Xi_N(\Upsilon_r)<0$. Then bisect the interval $[\Upsilon_l,\Upsilon_r]$ by computing the midpoint $\Upsilon_m=(\Upsilon_l+\Upsilon_r)/2$, replace Υ_l by Υ_m when $\Xi_N(\Upsilon_m)>0$, and replace Υ_r by Υ_m when $\Xi_N(\Upsilon_m)<0$. Return Υ_m if $\Xi_N(\Upsilon_m)=0$ or $|\Upsilon_r-\Upsilon_l|<\epsilon$. Corollary 1 proves that the doubling terminates, the bracket always contains Υ_N during the bisection, and the midpoint converges to Υ_N.

Lemma 2 (One-step map). *Fix $q\in(0,1)$ and $p\in(0,q)$, and define*

$$F_p(\theta)\triangleq\theta+\arccos\left(1-q-p\cos\theta\right).$$

Then:

1. *F_p is well defined on $\mathbb{R}$: the argument of the* arccos *lies in* $(1-2q,1) \subset (-1,1)$*, so* $F_p(\theta)-\theta \in (0,\pi)$*;*

2. *F_p is strictly increasing in θ;*

3. *for fixed θ with $\cos\theta < 0$, the map $p \mapsto F_p(\theta)$ is strictly decreasing.*

Proof. Write $u \triangleq 1-q-p\cos\theta$. Claim 1 follows from $|p\cos\theta| \le p < q$:

$$-1 < 1-2q < 1-q-p \le u \le 1-q+p < 1.$$

For claim 2, differentiating in θ gives

$$F_p'(\theta) = 1 - \frac{p\sin\theta}{\sqrt{1-u^2}},$$

so it suffices to prove $p^2\sin^2\theta < 1-u^2$: expanding $u = 1-q-p\cos\theta$, tedious but straightforward algebra gives $p^2\sin^2\theta + u^2 \le (p+1-q)^2 < 1$, the last inequality because $0 < p+1-q < 1$ by $p < q$. This proves claim 2. For claim 3, differentiating in p gives for $\cos\theta < 0$,

$$\frac{\partial F_p(\theta)}{\partial p} = \frac{\cos\theta}{\sqrt{1-u^2}} < 0.$$

□

By Lemma 2, the orbit in (5) which is written as $\theta_k(\Upsilon) = F_{p(\Upsilon)}\left(\theta_{k+1}(\Upsilon)\right)$ is defined for every $\Upsilon > 1$, each $\theta_k(\Upsilon)$ is a continuous function of Υ, and

$$\theta_1(\Upsilon) > \theta_2(\Upsilon) > \cdots > \theta_N(\Upsilon) = \arccos\left(-\sqrt{q}\right) > \frac{\pi}{2}.$$

Lemma 3 (Orbit comparison)**.** *Let $N \ge 2$ and $1 < \Upsilon < \Upsilon'$. If $\theta_1(\Upsilon) \le \pi$, then $\theta_k(\Upsilon') < \theta_k(\Upsilon)$ for $1 \le k \le N-1$.*

Proof. The hypothesis gives $\theta_k(\Upsilon) \in (\pi/2, \pi]$, hence $\cos\theta_k(\Upsilon) < 0$, for every $1 \le k \le N$. We induct downward on k. For the base case $k = N-1$, since $\cos\theta_N = -\sqrt{q} < 0$ and $p(\Upsilon') > p(\Upsilon)$, claim 3 of Lemma 2 gives

$$\theta_{N-1}(\Upsilon') = F_{p(\Upsilon')}\left(\theta_N\right) < F_{p(\Upsilon)}\left(\theta_N\right) = \theta_{N-1}(\Upsilon).$$

For the step from $k+1$ to $k \le N-2$, claims 2 and 3 of Lemma 2 give

$$\theta_k(\Upsilon') = F_{p(\Upsilon')}\left(\theta_{k+1}(\Upsilon')\right) < F_{p(\Upsilon')}\left(\theta_{k+1}(\Upsilon)\right) < F_{p(\Upsilon)}\left(\theta_{k+1}(\Upsilon)\right) = \theta_k(\Upsilon),$$

where the first inequality uses the induction hypothesis and the second uses $\cos\theta_{k+1}(\Upsilon) < 0$. □

Lemma 4 (Target angle)**.** *The target angle (4) is well defined, continuous, and strictly increasing on $(1,\infty)$, with $\vartheta(\Upsilon) \in (0,\pi)$ for every $\Upsilon > 1$, $\vartheta(\Upsilon) \to \arccos\left(\sqrt{q}\right)$ as $\Upsilon \downarrow 1$, and $\vartheta(\Upsilon) \to \pi$ as $\Upsilon \to \infty$.*

Proof. Write $c(\Upsilon)$ for the fraction in (4), and note that its denominator is $1+(1-q)R^2 = q+(1-q)\Upsilon^2$. Then $c(\Upsilon) < 1$, since the numerator is at most $\sqrt{q} < 1 < q+(1-q)\Upsilon^2$. Moreover, using $\Upsilon(\Upsilon-R) = \Upsilon/(\Upsilon+R)$ (from $\Upsilon^2 - R^2 = 1$), tedious but straightforward algebra writes $c(\Upsilon)+1$ as a ratio of positive quantities, so $c(\Upsilon) > -1$: hence $\vartheta(\Upsilon) = \arccos c(\Upsilon)$ is well defined in $(0,\pi)$, and it is continuous. The numerator is also strictly decreasing, since $\Upsilon/(\Upsilon+R) = 1/\left(1+R/\Upsilon\right)$ with $R/\Upsilon = \sqrt{1-\Upsilon^{-2}}$ strictly increasing, and the denominator is positive and strictly increasing; so c is strictly decreasing on $(1,\infty)$, and composing with the strictly decreasing arccos shows that ϑ is strictly increasing. The limits follow by substitution: as $\Upsilon \downarrow 1$, $R \to 0$ and $c \to \sqrt{q}$; as $\Upsilon \to \infty$, dividing the numerator and denominator of c by $(1-q)\Upsilon^2$ gives $c \to -1$. □

For $\Upsilon > 1$, call a collection of points an *admissible configuration at* Υ if it satisfies the ordering, positivity, and conditions of Lemma 1 with Υ_N replaced by Υ (and hence with $C = C(\Upsilon)$ and $R = R(\Upsilon)$).

Lemma 5 (Shooting equivalence)**.** *An admissible configuration at* $\Upsilon > 1$ *exists if and only if* $\Xi_N(\Upsilon) = 0$*. When it exists, it is unique.*

Proof. Suppose an admissible configuration exists at Υ. By the angle parametrization, its angles satisfy $\theta_N = \arccos\left(-\sqrt{q}\right)$ and the backward recurrence (5), so they coincide with the orbit $\theta_k(\Upsilon)$; in particular $\theta_1(\Upsilon) < \pi$. Moreover, the shooting equation (3) together with $a_1 > a_0$ places P_1 at the farther intersection, so $\cos\theta_1(\Upsilon) = \cos\vartheta(\Upsilon)$; both angles lie in $(0, \pi)$, hence $\theta_1(\Upsilon) = \vartheta(\Upsilon)$. Every coordinate is forced (P_0 and P_{N+1} by their formulas, P_k by $\theta_k(\Upsilon)$), which is the uniqueness claim.

Conversely, suppose $\theta_1(\Upsilon) = \vartheta(\Upsilon)$. Define $P_k \triangleq C + R\left(\cos\theta_k(\Upsilon), \sin\theta_k(\Upsilon)\right)$ for $1 \le k \le N$ and the endpoints by their formulas. Then all $\theta_k(\Upsilon)$ lie in $(0, \pi)$ because $0 < \theta_N \le \theta_k(\Upsilon) \le \theta_1(\Upsilon) = \vartheta(\Upsilon) < \pi$, so $b_k = R\sin\theta_k(\Upsilon) > 0$; the angles decrease strictly in k, so $a_1 < \cdots < a_N$; and $a_N = \Upsilon - \sqrt{q}\,R < \Upsilon = a_{N+1}$. The norm condition holds by construction. For the inner-product condition at $1 \le k \le N-1$, taking cosines in (5) gives $\cos\left(\theta_k - \theta_{k+1}\right) = 1 - q - p\cos\theta_{k+1} = 1 - q\,a_{k+1}/\Upsilon$, and $(P_k - C)\cdot(P_{k+1} - C) = R^2\cos\left(\theta_k - \theta_{k+1}\right)$ since both points lie on the circle; at $k = N$ the condition holds because $\sin\theta_N = \sqrt{1-q}$. At $k = 0$: the point $\left(a_+, \sqrt{1-q}\,R\,a_+\right)$ lies on the circle with positive ordinate, and P_1 is the point of the circle with the same abscissa $a_1 = \Upsilon + R\cos\vartheta(\Upsilon) = a_+$ and positive ordinate, so the two coincide; hence $\sin\theta_1(\Upsilon) = \sqrt{1-q}\,a_1$, which is (3) and, reversing the algebra that produced it, the inner-product condition at $k = 0$. Finally $a_0 = 1/\Upsilon < a_+ = a_1$. ☐

Proof of Lemma 1. Existence. We first show $\Xi_N(\Upsilon) < 0$ for all large Υ. Consider the boundary parameter $p = q$: the map $F_q(\theta) = \theta + \arccos\left(1 - q(1 + \cos\theta)\right)$ is still well defined, since $1 - q(1 + \cos\theta) \in [1 - 2q, 1]$. For $\theta \in (0, \pi)$,

$$\begin{aligned} F_q(\theta) < \pi &\Leftrightarrow \arccos\left(1 - q(1 + \cos\theta)\right) < \pi - \theta \\ &\Leftrightarrow 1 - q(1 + \cos\theta) > -\cos\theta \\ &\Leftrightarrow (1 - q)(1 + \cos\theta) > 0, \end{aligned}$$

which holds; so the orbit $\overline{\theta}_N \triangleq \theta_N, \overline{\theta}_k \triangleq F_q(\overline{\theta}_{k+1})$ satisfies $\overline{\theta}_1 < \pi$ by induction. Each backward step $(p, \theta) \mapsto F_p(\theta)$ is jointly continuous for $p \in [0, q]$ because the arccos argument stays in $[1 - 2q, 1]$; at the endpoint $p = 0$ it reads $F_0(\theta) = \theta + \arccos\left(1 - q\right)$. Hence $\theta_1(\Upsilon)$, which depends on Υ only through $p(\Upsilon)$, converges to $\overline{\theta}_1$ as $p(\Upsilon) \to q$, that is, as $\Upsilon \to \infty$. Combining with Lemma 4,

$$\lim_{\Upsilon \to \infty} \Xi_N(\Upsilon) = \overline{\theta}_1 - \pi < 0,$$

so $\Xi_N(b) < 0$ for some $b > 1$.

Next we produce a point where $\Xi_N > 0$. As $\Upsilon \downarrow 1$ we have $p(\Upsilon) \to 0$, so by the same joint continuity, now at the endpoint $p = 0$, $\theta_1(\Upsilon) \to \theta_N + (N-1)\arccos\left(1 - q\right)$, while $\vartheta(\Upsilon) \to \arccos\left(\sqrt{q}\right)$ by Lemma 4. Therefore

$$\lim_{\Upsilon \downarrow 1} \Xi_N(\Upsilon) = \theta_N + (N-1)\arccos\left(1 - q\right) - \arccos\left(\sqrt{q}\right) \ge \pi - 2\arccos\left(\sqrt{q}\right) = \arccos\left(1 - 2q\right) > 0,$$

where the middle inequality drops the nonnegative term $(N-1)\arccos\left(1-q\right)$ and uses $\theta_N = \arccos\left(-\sqrt{q}\right) = \pi - \arccos\left(\sqrt{q}\right)$, and the last equality holds because $\cos\left(\pi - 2\arccos\sqrt{q}\right) = -\left(2q - 1\right) = 1 - 2q$ and $\pi - 2\arccos\sqrt{q} \in (0, \pi)$. By continuity, $\Xi_N(a) > 0$ for some $a > 1$.

Hence $\Xi_N(a) > 0 > \Xi_N(b)$ with $a < b$ (enlarging b if necessary), so the intermediate value theorem yields $\Upsilon_N \in (a,b)$ with $\Xi_N(\Upsilon_N) = 0$. By Lemma 5, an admissible configuration exists at Υ_N.

Uniqueness. Suppose $\Upsilon < \Upsilon'$ both admit admissible points, so that $\theta_1(\Upsilon) = \vartheta(\Upsilon) < \pi$ and $\theta_1(\Upsilon') = \vartheta(\Upsilon')$ by Lemma 5. For $N = 1$ the orbit is the constant $\arccos\left(-\sqrt{q}\right)$, and the strict monotonicity of ϑ makes the two equalities incompatible. For $N \geq 2$, Lemma 3 applies at Υ (its hypothesis $\theta_1(\Upsilon) \leq \pi$ holds) and gives, with Lemma 4,

$$\theta_1(\Upsilon') < \theta_1(\Upsilon) = \vartheta(\Upsilon) < \vartheta(\Upsilon') = \theta_1(\Upsilon'),$$

a contradiction. Hence the admissible Υ is unique, and Lemma 5 shows that the points are uniquely determined by it. $\square$

The same objects also give a sign characterization of the residual that makes the numerical evaluation of Υ_N straightforward.

Corollary 1 (Bisection correctness)**.** *Let Υ_N be as in Lemma 1. Then the residual Ξ_N defined above satisfies $\Xi_N(\Upsilon) > 0$ for $1 < \Upsilon < \Upsilon_N$ and $\Xi_N(\Upsilon) < 0$ for $\Upsilon > \Upsilon_N$. Consequently both update rules of the bisection described above are correct, the doubling of the right endpoint terminates and produces a bracket containing Υ_N, the invariant $\Upsilon_l < \Upsilon_N < \Upsilon_r$ is maintained, and the midpoints converge to Υ_N.*

Proof. By Lemma 5, $\theta_1(\Upsilon_N) = \vartheta(\Upsilon_N) < \pi$, so $\Xi_N(\Upsilon_N) = 0$. For $\Upsilon > \Upsilon_N$, Lemma 3 applies at Υ_N (its hypothesis $\theta_1(\Upsilon_N) \leq \pi$ holds) and gives, with the strictly increasing target angle of Lemma 4,

$$\Xi_N(\Upsilon) = \theta_1(\Upsilon) - \vartheta(\Upsilon) < \theta_1(\Upsilon_N) - \vartheta(\Upsilon_N) = 0.$$

For $1 < \Upsilon < \Upsilon_N$, either $\theta_1(\Upsilon) \geq \pi > \vartheta(\Upsilon)$ and $\Xi_N(\Upsilon) > 0$ directly, or $\theta_1(\Upsilon) < \pi$ and Lemma 3, now applied at Υ, gives in the same way $\Xi_N(\Upsilon) > \Xi_N(\Upsilon_N) = 0$. (For $N = 1$ the orbit is the constant $\arccos\left(-\sqrt{q}\right)$ and both inequalities hold through the target angle alone.) For the bisection, the doubling terminates, since $\Xi_N < 0$ once Υ_r exceeds Υ_N, producing a bracket $\Upsilon_l = 1 < \Upsilon_N < \Upsilon_r$; by the sign characterization each update rule fires exactly on its side of Υ_N, so each update preserves the invariant $\Upsilon_l < \Upsilon_N < \Upsilon_r$ while halving the bracket, and the midpoints converge to Υ_N. $\square$

Lemma 6 (Symmetry of the construction)**.** *For $0 \leq k \leq N+1$,*

$$\begin{aligned} a_k a_{N+1-k} &= 1, \\ a_k b_{N+1-k} &= b_k. \end{aligned} \tag{6}$$

Proof. The circle equation is equivalent to $a^2 + b^2 - 2\Upsilon_N a + 1 = 0$, and the map $T(a,b) = (1/a, b/a)$ maps this circle to itself, with $T(P_0) = P_{N+1}$. Moreover, if $P = (a,b)$ and $P' = (a', b')$ satisfy

$$(P - C) \cdot (P' - C) = R_N^2 \left(1 - \frac{qa'}{\Upsilon_N}\right),$$

then direct expansion gives

$$(T(P') - C) \cdot (T(P) - C) = \frac{(P - C) \cdot (P' - C) + R_N^2(aa' - 1)}{aa'} = R_N^2 \left(1 - \frac{q}{\Upsilon_N a}\right).$$

This is the recurrence for the reversed pair, since the first coordinate of $T(P)$ is $1/a$. Hence $\widetilde{P}_k \triangleq T(P_{N+1-k})$ satisfies the same endpoint conditions and recurrence as P_k. Its first coordinates are strictly increasing and its second coordinates are positive, so uniqueness in Lemma 1 gives $T(P_k) = P_{N+1-k}$. Comparing coordinates proves (6). $\square$

The construction also gives the explicit relaxation used in Theorem 1.

Lemma 7. *Let Υ_N be as in Lemma 1. Then, for every $N \geq 1$,*

$$\frac{1}{\Upsilon_N^2} < 4(1-\sqrt{q})^{2N}.$$

Proof. For the construction angles θ_k, define $t_k \triangleq \cot(\theta_k/2)$ for $1 \leq k \leq N$. The half-angle formula gives

$$t_k = \frac{\sin\theta_k}{1-\cos\theta_k} = \frac{b_k}{\Upsilon_N + R_N - a_k}.$$

By the symmetry of Lemma 6, $a_1 a_N = 1$ and $b_1 = a_1 b_N$; substituting these together with $a_N = \Upsilon_N - \sqrt{q}\,R_N$ and $b_N = \sqrt{1-q}\,R_N$ into the half-angle formula, tedious but straightforward algebra gives

$$\begin{aligned} t_1 &= \frac{\sqrt{1-q}}{1-\sqrt{q}}\,\frac{1}{\Upsilon_N + R_N}, \\ t_N &= \frac{\sqrt{1-q}}{1+\sqrt{q}}. \end{aligned}$$

For $1 \leq k \leq N-1$, the inner-product condition, $\cos\theta_{k+1} < 0$, and $R_N < \Upsilon_N$ give

$$\begin{aligned} \cos(\theta_k - \theta_{k+1}) &= 1 - q\left(1 + \frac{R_N}{\Upsilon_N}\cos\theta_{k+1}\right) \\ &< 1 - q(1+\cos\theta_{k+1}) = 1 - 2q\cos^2\left(\frac{\theta_{k+1}}{2}\right), \end{aligned}$$

and hence, taking half-angle sines, expanding $\sin((\theta_k - \theta_{k+1})/2)$, and dividing by $\sin(\theta_k/2)\sin(\theta_{k+1}/2)$ (tedious but straightforward algebra), $t_k < (1-\sqrt{q})t_{k+1}$ for $1 \leq k \leq N-1$, which gives $t_1 \leq (1-\sqrt{q})^{N-1} t_N$.

Combining the contraction with these boundary values,

$$\frac{1}{2\Upsilon_N} < \frac{1}{\Upsilon_N + R_N} = t_1 t_N \leq t_N^2 (1-\sqrt{q})^{N-1} = \frac{(1-\sqrt{q})^N}{1+\sqrt{q}}.$$

It follows that $1/\Upsilon_N^2 < \left(2/(1+\sqrt{q})\right)^2 (1-\sqrt{q})^{2N} < 4(1-\sqrt{q})^{2N}$, which completes the proof. $\square$

Two further identities of the construction. The equivalence argument of Section 8 consumes two identities of the construction that we record here, both consequences of the conditions (1) and the symmetry (6). For the first, define

$$s_k \triangleq \frac{\sqrt{q}\,a_{k+1}\phi_{N-k}}{\Upsilon_N}, \qquad 1 \leq k \leq N, \tag{7}$$

where the coefficients ϕ_j are those of ITEM-f, and recall the radius vectors $\widehat{r}_k \triangleq P_k - C$ for $0 \leq k \leq N+1$.

Lemma 8 (Signed areas)**.** *For $1 \leq k \leq N$,*

$$\det\left(\widehat{r}_{k+1}, \widehat{r}_k\right) = R_N^2\, s_k.$$

Proof. First let $1 \le k \le N-1$, so that both P_k and P_{k+1} lie on the circle. Writing $\widehat{r}_j = R_N(\cos\theta_j, \sin\theta_j)$ with the construction angles $\theta_1 > \cdots > \theta_N$ in $(0,\pi)$,

$$\det\left(\widehat{r}_{k+1}, \widehat{r}_k\right) = R_N^2\left(\cos\theta_{k+1}\sin\theta_k - \sin\theta_{k+1}\cos\theta_k\right) = R_N^2\sin\left(\theta_k - \theta_{k+1}\right).$$

The inner-product condition (second line of (1)) reads $\cos(\theta_k - \theta_{k+1}) = 1 - q\,a_{k+1}/\Upsilon_N$, so

$$\sin^2\left(\theta_k - \theta_{k+1}\right) = 1 - \left(1 - \frac{q\,a_{k+1}}{\Upsilon_N}\right)^2 = \frac{q\,a_{k+1}}{\Upsilon_N^2}\left(2\Upsilon_N - q\,a_{k+1}\right).$$

On the other hand, the symmetry (6) gives $a_{N-k}a_{k+1} = 1$, and since both k and $N-k$ lie in $\{1,\dots,N-1\}$, the definition of ϕ_{N-k} gives

$$\phi_{N-k}^2 = 2\Upsilon_N a_{N-k} - q = \frac{2\Upsilon_N - q\,a_{k+1}}{a_{k+1}},$$

whence

$$s_k^2 = \frac{q\,a_{k+1}^2\,\phi_{N-k}^2}{\Upsilon_N^2} = \frac{q\,a_{k+1}}{\Upsilon_N^2}\left(2\Upsilon_N - q\,a_{k+1}\right) = \sin^2\left(\theta_k - \theta_{k+1}\right).$$

Both quantities are positive: $0 < \theta_k - \theta_{k+1} < \pi$ by claim 1 of Lemma 2, and $s_k > 0$ since $a_{k+1} > 0$ and $\phi_{N-k} > 0$ (indeed $2\Upsilon_N a_j \ge 2\Upsilon_N a_0 = 2 > q$ for every j). Taking positive square roots proves the interior case.

For $k = N$, the point P_{N+1} is not on the circle, and we compute directly. As established in the proof of Lemma 5, $a_N = \Upsilon_N - \sqrt{q}\,R_N$, and $b_N = b_{N+1} = \sqrt{1-q}\,R_N$; hence

$$\widehat{r}_N = \begin{pmatrix} -\sqrt{q}\,R_N \\ \sqrt{1-q}\,R_N \end{pmatrix}, \qquad \widehat{r}_{N+1} = \begin{pmatrix} 0 \\ \sqrt{1-q}\,R_N \end{pmatrix},$$

and $\det(\widehat{r}_{N+1}, \widehat{r}_N) = R_N^2\sqrt{q(1-q)} = R_N^2 s_N$, the last equality because $a_{N+1} = \Upsilon_N$ and $\phi_0 = \sqrt{1-q}$ give $s_N = \sqrt{q(1-q)}$ in (7). □

Lemma 9 (Product identity for the momentum coefficients)**.** *For $1 \le k \le N-1$,*

$$\phi_k\,\phi_{N-k} = \Upsilon_N\left(1 - q + \frac{a_k}{a_{k+1}}\right).$$

With the convention $\phi_N = \big((1-q)\Upsilon_N + a_N\big)/\sqrt{1-q}$ of ITEM-f, the identity also holds at $k=0$ and $k=N$.

Proof. Let $1 \le k \le N-1$, write $D_j \triangleq a_j - \Upsilon_N$, and abbreviate $R = R_N$, $\Upsilon = \Upsilon_N$. Both P_k and P_{k+1} lie on the circle, so $b_j^2 = R^2 - D_j^2$ for $j \in \{k, k+1\}$, and the inner-product condition reads

$$b_k b_{k+1} = R^2\left(1 - \frac{q\,a_{k+1}}{\Upsilon}\right) - D_k D_{k+1}.$$

Squaring this equation (both sides are positive, being equal to $b_k b_{k+1} > 0$), substituting $b_k^2 b_{k+1}^2 = (R^2 - D_k^2)(R^2 - D_{k+1}^2)$, and using the algebraic identity

$$\left(R^2 - D_k D_{k+1}\right)^2 - \left(R^2 - D_k^2\right)\left(R^2 - D_{k+1}^2\right) = R^2\left(D_k - D_{k+1}\right)^2$$

together with $R^2 - D_k D_{k+1} = \Upsilon(a_k + a_{k+1}) - a_k a_{k+1} - 1$ (which uses $R^2 = \Upsilon^2 - 1$), we obtain, after multiplying through by Υ^2/R^2, the consecutive-pair relation

$$\Upsilon^2\left(a_k - a_{k+1}\right)^2 = 2q\Upsilon a_{k+1}\left(\Upsilon a_k + \Upsilon a_{k+1} - a_k a_{k+1} - 1\right) - q^2\left(\Upsilon^2 - 1\right)a_{k+1}^2. \tag{8}$$

Expanding both sides and comparing term by term, tedious but straightforward algebra shows that

$$\Upsilon^2\big((1-q)a_{k+1}+a_k\big)^2 - a_{k+1}\,(2\Upsilon a_k - q)\,(2\Upsilon - q\,a_{k+1})$$

equals the left-hand side of (8) minus its right-hand side, and therefore vanishes. Since $\phi_k^2 = 2\Upsilon a_k - q$ and, as in the proof of Lemma 8, $\phi_{N-k}^2 = (2\Upsilon - q\,a_{k+1})\,/a_{k+1}$, dividing by a_{k+1}^2 gives

$$\phi_k^2\,\phi_{N-k}^2 = \Upsilon^2\left(1-q+\frac{a_k}{a_{k+1}}\right)^2.$$

All of ϕ_k, ϕ_{N-k}, and $\Upsilon\,(1-q+a_k/a_{k+1})$ are positive, so taking positive square roots proves the interior case.

At $k=0$, the convention for ϕ_N and the symmetry (6) give

$$\phi_0\,\phi_N = \sqrt{1-q}\cdot\frac{(1-q)\Upsilon+a_N}{\sqrt{1-q}} = (1-q)\Upsilon + a_N = (1-q)\Upsilon + \frac{1}{a_1} = \Upsilon\left(1-q+\frac{a_0}{a_1}\right),$$

using $a_1 a_N = 1$ and $\Upsilon a_0 = 1$. The case $k=N$ is the same identity with the two factors exchanged, since $\Upsilon\,(1-q+a_N/a_{N+1}) = (1-q)\Upsilon + a_N$ by $a_{N+1} = \Upsilon$. □

5 PEP formulation of the worst case

We now formulate the worst-case performance of a fixed-step first-order method for this setup as an optimization problem and reduce it to a convex SDP whose dual carries convergence certificates, following the performance estimation methodology [3, 6].

A *fixed-step first-order method* (FSFOM) with N steps produces its iterates from a starting point $x_0 \in \mathbb{R}^d$ with

$$x_i = x_{i-1} - \frac{1}{L}\sum_{j=0}^{i-1} h_{i,j}\nabla f(x_j), \quad 1 \le i \le N, \tag{9}$$

with stepsizes $\{h_{i,j}\}_{0\le j<i\le N}$ fixed in advance, independently of f, normalized here by L. We rate a method by its worst-case function-value suboptimality after N steps, relative to the initial suboptimality. Since $\mathcal{F}_{\mu,L}$ is invariant under translation of the variable and shift of the function value, we set $x_\star = 0$, $f_\star = 0$, $\nabla f(x_\star) = 0$ and normalize the initial gap by $f(x_0) - f_\star \le 1$. The worst-case performance of an FSFOM M is

$$\mathcal{R}(M) = \left(\begin{array}{ll} \text{maximize} & f(x_N) - f_\star \\ \text{subject to} & f \in \mathcal{F}_{\mu,L}, \\ & \{x_i\}_{1\le i\le N} \text{ generated from } x_0 \text{ by } M, \\ & f(x_0) - f_\star \le 1, \\ & x_\star = 0,\ f_\star = 0,\ \nabla f(x_\star) = 0 \end{array}\right), \tag{$\mathcal{O}_f^{\mathrm{in}}$}$$

where the decision variables are the function f and the iterates $x_0,\dots,x_N$. Any upper bound on $\mathcal{R}(M)$ is a convergence guarantee for M; producing such a bound for the stepsizes of ITEM-f is what the rest of this document does.

Following [5], we pass to a transformed function. By scale invariance [6, §3.5] fix $L=1$, so $\mu = q$. For $f \in \mathcal{F}_{q,1}$ with $x_\star = 0$, $f_\star = 0$, the shifted function $\tilde f \triangleq f - (q/2)\,\|\cdot\|^2$ belongs to $\mathcal{F}_{0,1-q}$, with gradients $\tilde g_i = \nabla f(x_i) - q\,x_i$ and $\tilde g_\star = 0$; the original suboptimality is recovered from the transformed values $\tilde f_i = \tilde f(x_i)$ through

$$f(x_i) - f_\star = \tilde f_i + \tfrac{q}{2}\,\|x_i\|^2. \tag{10}$$

Re-expressing (9) through $\tilde{g}_i$, the iterates take the form

$$x_i = \Big(1 - q\sum_{j=0}^{i-1}\alpha_{i,j}\Big)x_0 - \sum_{j=0}^{i-1}\alpha_{i,j}\,\tilde{g}_j, \quad 1 \le i \le N, \tag{11}$$

where the transformed stepsizes $\{\alpha_{i,j}\}_{0\le j<i\le N}$ is defined from the original stepsizes $\{h_{i,j}\}_{0\le j<i\le N}$ through the reparametrization [5, §3.2]

$$\alpha_{i,j} \triangleq \begin{cases} h_{i,i-1}, & j = i-1, \\ h_{i,j} + \alpha_{i-1,j} - q\displaystyle\sum_{k=j+1}^{i-1} h_{i,k}\,\alpha_{k,j}, & 0 \le j \le i-2, \end{cases} \tag{12}$$

with inverse transformation

$$h_{i,j} = \begin{cases} \alpha_{i,i-1}, & j = i-1, \\ \alpha_{i,j} - \alpha_{i-1,j} + q\displaystyle\sum_{k=j+1}^{i-1} h_{i,k}\,\alpha_{k,j}, & 0 \le j \le i-2. \end{cases} \tag{13}$$

This correspondence is invertible, so we take $\{\alpha_{i,j}\}_{0\le j<i\le N}$ as the design variables.

The transformed function is discretized by the following interpolation theorem.

Theorem 2 ($\mathcal{F}_{0,L}$-interpolation [6, Corollary 1])**.** *Let I be an index set, and let $\{(x_i, g_i, f_i)\}_{i\in I} \subseteq \mathbb{R}^d \times \mathbb{R}^d \times \mathbb{R}$. Let $L > 0$. There exists $f \in \mathcal{F}_{0,L}$ satisfying $f(x_i) = f_i$ and $g_i \in \partial f(x_i)$ for all $i \in I$ if and only if*

$$f_i \ge f_j + \langle g_j, x_i - x_j\rangle + \frac{1}{2L}\left\|g_i - g_j\right\|^2$$

for all $i, j \in I$.

Applied to $\tilde{f} \in \mathcal{F}_{0,1-q}$, with $1-q$ in place of L, these inequalities discretize the transformed problem at the iterates.

Collect the data into $P = [\,x_0 \mid \tilde{g}_0 \mid \cdots \mid \tilde{g}_N\,] \in \mathbb{R}^{d\times(N+2)}$, $G = P^\top P \in \mathbb{S}_+^{N+2}$, and $F = [\,\tilde{f}_0 \mid \cdots \mid \tilde{f}_N\,]$. With the selectors $\mathbf{x}_0 = e_1$, $\mathbf{g}_i = e_{i+2}$ (so $\tilde{g}_i = P\mathbf{g}_i$), $\mathbf{f}_i = e_{i+1}$, $\mathbf{x}_\star = \mathbf{g}_\star = 0$, and $\mathbf{f}_\star = 0$, define the iterate selectors by mirroring the coefficients of (11),

$$\mathbf{x}_i \triangleq \Big(1 - q\sum_{j=0}^{i-1}\alpha_{i,j}\Big)\mathbf{x}_0 - \sum_{j=0}^{i-1}\alpha_{i,j}\,\mathbf{g}_j, \quad 1 \le i \le N, \tag{14}$$

so that $x_i = P\mathbf{x}_i$ by (11), and define, for $i, j \in I_N^\star$,

$$\begin{aligned} A_{i,j}(\alpha) &= \mathbf{g}_j \odot (\mathbf{x}_i - \mathbf{x}_j) \in \mathbb{S}^{N+2}, \\ B_{i,j}(\alpha) &= (\mathbf{x}_i - \mathbf{x}_j) \odot (\mathbf{x}_i - \mathbf{x}_j) \in \mathbb{S}_+^{N+2}, \\ C_{i,j} &= (\mathbf{g}_i - \mathbf{g}_j) \odot (\mathbf{g}_i - \mathbf{g}_j) \in \mathbb{S}_+^{N+2}. \end{aligned} \tag{15}$$

By (14) $\mathbf{x}_i$ is affine in α, so $A_{i,j}$ is affine and $B_{i,j}$ quadratic in α, while $C_{i,j}$ is constant. By (10) the objective is $f(x_N) - f_\star = F\mathbf{f}_N + (q/2)\mathbf{tr}(GB_{N,\star})$ and the initial gap is $f(x_0) - f_\star = F\mathbf{f}_0 + (q/2)\mathbf{tr}(GB_{0,\star})$. Lifting to (F, G) is exact at the price of the rank constraint $\mathbf{rank}\, G \le d$; we drop the rank constraint, which can only enlarge the feasible set, so the optimal value of the resulting problem upper-bounds $\mathcal{R}(M)$ in every dimension d, and the relaxation is exact when $d \geq N+2$ [6, §3.2]. Only the upper-bound direction is used below, so the guarantee we prove is dimension-free. The relaxed problem is the convex SDP

$$\mathcal{R}(M) \leq \left(\begin{array}{l} \text{maximize} \quad F\mathbf{f}_N + \frac{q}{2}\mathbf{tr} G B_{N,\star} \\ \text{subject to} \\ F(\mathbf{f}_j - \mathbf{f}_i) + \mathbf{tr} G\big[A_{i,j}(\alpha) + \frac{1}{2(1-q)} C_{i,j}\big] \leq 0, \quad i,j \in I_N^\star : i \neq j, \quad \triangleright \text{dual var. } \lambda_{i,j} \geq 0 \\ F\mathbf{f}_0 + \frac{q}{2}\mathbf{tr} G B_{0,\star} \leq 1, \quad \triangleright \text{dual var. } \nu \geq 0 \\ G \succeq 0, \quad \triangleright \text{dual var. } Z \succeq 0 \end{array} \right), \tag{16}$$

with decision variables $F \in \mathbb{R}^{1\times(N+1)}$ and $G \in \mathbb{S}^{N+2}$, and the indicated dual variables. Taking the dual gives

$$\overline{\mathcal{R}}(M) = \left(\begin{array}{l} \text{minimize} \quad \nu \\ \text{subject to} \\ \mathbf{f}_N - \nu\, \mathbf{f}_0 + \sum_{i,j \in I_N^\star : i \neq j} \lambda_{i,j}\, (\mathbf{f}_i - \mathbf{f}_j) = 0, \\ \frac{q}{2}\big(\nu B_{0,\star} - B_{N,\star}(\alpha)\big) + \sum_{i,j \in I_N^\star : i\neq j} \lambda_{i,j} \big[A_{i,j}(\alpha) + \frac{1}{2(1-q)} C_{i,j}\big] = Z, \\ Z \succeq 0, \quad \nu \geq 0, \quad \lambda_{i,j} \geq 0 \ \ (i,j \in I_N^\star : i \neq j) \end{array} \right), \tag{17}$$

with decision variables $\nu \in \mathbb{R}$, $\lambda = \{\lambda_{i,j}\}$, and $Z \in \mathbb{S}^{N+2}$, the *inner-dual variables.* By weak duality, every feasible point of (17) upper-bounds the SDP value in (16), and therefore $\mathcal{R}(M) \leq \overline{\mathcal{R}}(M)$. Consequently, to prove a convergence guarantee it suffices to exhibit a single feasible point of (17): its objective value ν is a valid upper bound on the worst-case function-value ratio. Strong duality is never needed for this direction.

6 The analytic certificate

We now state the analytic solution: transformed stepsizes together with inner-dual variables that, as Section 7 proves, form a feasible point of (17). How these expressions were originally found is described in Remark 1; the verification below is entirely independent of that discovery process. The expressions are stated in terms of the construction of Lemma 1 (at $L = 1$, hence $\mu = q$). Introduce the radius vectors, their edge differences, and the normalized radius vectors

$$\begin{aligned} \widehat{r}_k &\triangleq P_k - C, && 0 \leq k \leq N+1, \\ \Delta\widehat{r}_k &\triangleq \widehat{r}_{k+1} - \widehat{r}_k, && 0 \leq k \leq N, \\ \widehat{u}_k &\triangleq \frac{\widehat{r}_k}{a_k}, && 1 \leq k \leq N+1. \end{aligned} \tag{18}$$

For the transformed stepsizes,

$$\alpha_{i,j} = c_\alpha\, \widehat{u}_{i+1} \cdot \Delta\widehat{r}_j, \quad 0 \leq j < i \leq N, \tag{19}$$

where $c_\alpha \triangleq \Upsilon_N/(q(1-q)R_N^2)$.

For inner-dual variables of (17) for the FSFOM M given by (19), $\nu = 1/\Upsilon_N^2$ and

$$\lambda = \begin{cases} \lambda_{i-1,i} = \dfrac{a_i}{\Upsilon_N}, \quad 1 \leq i \leq N, \\ \lambda_{\star,i} = \dfrac{a_{i+1} - a_i}{\Upsilon_N}, \quad 0 \leq i \leq N, \\ \lambda_{i,j} = 0, \quad \text{otherwise}, \end{cases} \tag{20}$$

and

$$Z = \frac{q}{2}\big(\nu B_{0,\star} - B_{N,\star}(\alpha)\big) + \sum_{i,j\in I_N^\star : i\neq j} \lambda_{i,j}\big[A_{i,j}(\alpha) + \frac{1}{2(1-q)}C_{i,j}\big]. \tag{21}$$

7 Feasibility of the certificate

In this section we prove that the proposed dual variables $\nu = 1/\Upsilon_N^2$, λ of (20), and the slack Z of (21) are feasible for the dual problem (17).

The argument is organized as follows. We first record the algebraic identities of the geometric construction (Lemmas 10–11). Feasibility of (17) amounts to three conditions, which we check in turn: **(1)** nonnegativity of the multipliers, **(2)** the linear function-value equation, and **(3)** positive semidefiniteness of the slack Z, which reduces to a single matrix identity (Lemma 12).

Auxiliary identities of the construction. For this proof, recall the radius vectors, their edge differences, and the normalized radius vectors defined in (18). Denote $\Delta a_k \triangleq a_{k+1} - a_k$ and $\Delta b_k \triangleq b_{k+1} - b_k$.

The second line of (1) at $k = N$, with the endpoint $a_{N+1} = \Upsilon_N$, gives $\widehat{r}_N \cdot \widehat{r}_{N+1} = b_N b_{N+1} = (1-q)R_N^2 = b_{N+1}^2$, hence $b_N = b_{N+1} = \sqrt{1-q}\,R_N$. (Here the first coordinate of $\widehat{r}_{N+1}$ is $a_{N+1} - \Upsilon_N = 0$, so the dot product reduces to $b_N b_{N+1}$, while $b_{N+1} = \sqrt{1-q}\,R_N$ is read off the endpoint P_{N+1} in the last line of (1), which gives the final equality.) The following identities follow from the conditions (1).

Lemma 10. *For $0 \le k \le N$, normalized radius vectors $\widehat{u}_k = \widehat{r}_k/a_k$ and edge vectors $\Delta\widehat{r}_k = \widehat{r}_{k+1} - \widehat{r}_k$ from the construction satisfy the following:*

$$\frac{\Upsilon_N}{qR_N^2}\,\|\Delta\widehat{r}_k\|^2 = \begin{cases} 2a_1 - a_0, & k = 0,\\ 2a_{k+1}, & 1 \le k \le N-1,\\ a_{N+1}, & k = N. \end{cases} \tag{22}$$

$$\frac{\Upsilon_N}{qR_N^2}\widehat{u}_{k+1}\cdot\Delta\widehat{r}_k = \begin{cases} 1, & 0 \le k \le N-1,\\ 0, & k = N. \end{cases} \tag{23}$$

Proof. For $1 \le k \le N-1$, expanding the square gives $\|\Delta\widehat{r}_k\|^2 = 2R_N^2 - 2\widehat{r}_k\cdot\widehat{r}_{k+1} = (2qR_N^2/\Upsilon_N)a_{k+1}$ by the second line of (1). For $k = 0$, $\|\widehat{r}_0\|^2 = R_N^2(1 - q/\Upsilon_N^2)$ and the recurrence at $k = 0$ give $\|\Delta\widehat{r}_0\|^2 = (qR_N^2/\Upsilon_N)(2a_1 - a_0)$. For $k = N$, $\|\widehat{r}_{N+1}\|^2 = (1-q)R_N^2$ gives $\|\Delta\widehat{r}_N\|^2 = qR_N^2 = (qR_N^2/\Upsilon_N)a_{N+1}$.

For the second equation, since $\Delta\widehat{r}_k = \widehat{r}_{k+1} - \widehat{r}_k$ and $\|\widehat{r}_{k+1}\|^2 = R_N^2$ for $0 \le k \le N-1$, $\Delta\widehat{r}_k\cdot\widehat{r}_{k+1} = \|\widehat{r}_{k+1}\|^2 - \widehat{r}_k\cdot\widehat{r}_{k+1} = (qR_N^2/\Upsilon_N)a_{k+1}$, while $\Delta\widehat{r}_N\cdot\widehat{r}_{N+1} = \|\widehat{r}_{N+1}\|^2 - \widehat{r}_N\cdot\widehat{r}_{N+1} = 0$. □

We also use the row-sum identity for the transformed stepsizes (19).

Lemma 11. *For $1 \le i \le N$,*

$$1 - q\sum_{j=0}^{i-1}\alpha_{i,j} = \frac{b_{N-i}}{b_N}.$$

Proof. Since $\sum_{j=0}^{i-1}\Delta\widehat{r}_j = \widehat{r}_i - \widehat{r}_0$ and $\widehat{u}_{i+1} = \widehat{r}_{i+1}/a_{i+1}$, (19) gives $\sum_{j=0}^{i-1}\alpha_{i,j} = \Upsilon_N/(qb_N^2 a_{i+1})\,\widehat{r}_{i+1}\cdot(\widehat{r}_i - \widehat{r}_0)$. Using $\widehat{r}_0 = (-R_N^2/\Upsilon_N,\, b_N/\Upsilon_N)^\top$, the second line of (1), and $b_N^2 = (1-q)R_N^2$,

$$\widehat{r}_{i+1}\cdot(\widehat{r}_i - \widehat{r}_0) = \frac{b_N^2}{\Upsilon_N}a_{i+1} - \frac{b_{i+1}b_N}{\Upsilon_N},$$

so $1 - q\sum_{j=0}^{i-1}\alpha_{i,j} = b_{i+1}/(a_{i+1}b_N) = b_{N-i}/b_N$, the last step by (6). $\square$

By the selector definition (14) and Lemma 11, the iterate selectors admit the representation

$$\mathbf{x}_k = \frac{b_{N-k}}{b_N}\,\mathbf{x}_0 - \sum_{i=0}^{k-1}\alpha_{k,i}\,\mathbf{g}_i, \quad 0 \le k \le N, \tag{24}$$

where $\mathbf{g}_i$ selects the transformed gradient $\tilde{g}_i$.

Feasibility of the certificate. We prove the feasibility by checking the constraints of (17). The second constraint holds by the definition of Z in (21), so we check the other three constraints.

1. Nonnegativity of the dual multipliers ν and λ.

Clearly $\nu = 1/\Upsilon_N^2 > 0$. Since $1/\Upsilon_N = a_0 < a_1 < \cdots < a_N < a_{N+1} = \Upsilon_N$, the multipliers $\lambda_{i-1,i} = a_i/\Upsilon_N$, $1 \le i \le N$, and $\lambda_{\star,i} = \Delta a_i/\Upsilon_N$, $0 \le i \le N$, are nonnegative, and all other entries vanish. Hence $\lambda \ge 0$.

2. The linear constraint $\mathbf{f}_N - \nu\,\mathbf{f}_0 + \sum_{i,j\in I_N^\star : i\neq j}\lambda_{i,j}\,(\mathbf{f}_i - \mathbf{f}_j) = \mathbf{0}$.

Substituting the dual multipliers (20) and $\nu = 1/\Upsilon_N^2$, the linear constraint becomes

$$\mathbf{f}_N - \frac{1}{\Upsilon_N^2}\mathbf{f}_0 + \sum_{i=1}^{N}\frac{a_i}{\Upsilon_N}(\mathbf{f}_{i-1} - \mathbf{f}_i) - \sum_{i=0}^{N}\frac{\Delta a_i}{\Upsilon_N}\mathbf{f}_i = \mathbf{0}.$$

Since the selectors $\mathbf{f}_0, \dots, \mathbf{f}_N$ are linearly independent, this holds if and only if the coefficient of each $\mathbf{f}_i$ vanishes. The coefficient of $\mathbf{f}_0$ is $-1/\Upsilon_N^2 + a_1/\Upsilon_N - (a_1 - a_0)/\Upsilon_N = -1/\Upsilon_N^2 + a_0/\Upsilon_N = 0$ since $a_0 = 1/\Upsilon_N$; the coefficient of $\mathbf{f}_i$ for $1 \le i \le N-1$ is $-a_i/\Upsilon_N + a_{i+1}/\Upsilon_N - (a_{i+1} - a_i)/\Upsilon_N = 0$; and the coefficient of $\mathbf{f}_N$ is $1 - a_N/\Upsilon_N - (a_{N+1} - a_N)/\Upsilon_N = 1 - a_{N+1}/\Upsilon_N = 0$ since $a_{N+1} = \Upsilon_N$.

3. Positive semidefiniteness of Z.

Multiplying the slack (21) by $2\Upsilon_N^2$ and substituting $\nu = 1/\Upsilon_N^2$ and the multipliers (20) (so that $2\Upsilon_N^2\lambda_{i-1,i} = 2\Upsilon_N a_i$ and $2\Upsilon_N^2\lambda_{\star,i} = 2\Upsilon_N\Delta a_i$) turns $2\Upsilon_N^2 Z$ into the left-hand side of (25) below; hence $Z \succeq 0$ follows from Lemma 12.

Lemma 12. *With $L = 1$ and $\mathbf{x}_\star = 0$, the method and the multipliers (20) satisfy*

$$\begin{aligned} &qB_{0,\star} - q\Upsilon_N^2 B_{N,\star} + 2\Upsilon_N\sum_{i=1}^{N} a_i\Big(A_{i-1,i} + \tfrac{1}{2(1-q)}C_{i-1,i}\Big) \\ &\quad + 2\Upsilon_N\sum_{i=0}^{N}\Delta a_i\Big(A_{\star,i} + \tfrac{1}{2(1-q)}C_{\star,i}\Big) = \frac{\Upsilon_N^2}{q(1-q)(\Upsilon_N^2 - 1)}\,\mathbf{z}\mathbf{z}^\top, \end{aligned} \tag{25}$$

where $\mathbf{z} = (0, \Delta a_0, \Delta a_1, \dots, \Delta a_N)^\top$. Consequently $Z = \frac{1}{2q(1-q)(\Upsilon_N^2-1)}\,\mathbf{z}\mathbf{z}^\top \succeq 0$.

Proof. We compute the left-hand side of (25), denoted LHS, using the iterate representation (24) and the *extended stepsizes*

$$\bar{\alpha}_{i,j} \triangleq c_\alpha\,\widehat{u}_{i+1}\cdot\Delta\widehat{r}_j, \quad 0 \le j \le i \le N,$$

which by (23) take the values $\bar{\alpha}_{i,j} = \alpha_{i,j}$ for $j < i$, $\bar{\alpha}_{i,i} = 1/(1-q)$ for $i < N$, and $\bar{\alpha}_{N,N} = 0$. They satisfy the forward-difference identity

$$a_{i+1}\bar{\alpha}_{i,j} - a_i\bar{\alpha}_{i-1,j} = c_\alpha\,\Delta\widehat{r}_i\cdot\Delta\widehat{r}_j, \tag{26}$$

and, since $\widehat{r}_{N+1} = (0, b_N)^\top$ and $a_{N+1} = \Upsilon_N$, the endpoint identity

$$q\,\bar{\alpha}_{N,i} = \frac{\Delta b_i}{b_N}, \quad 0 \le i \le N. \tag{27}$$

Denote $W = \sum_{i=0}^{N} \bar{\alpha}_{N,i}\mathbf{g}_i$. Taking $k = N$ in (24), with $b_0 = a_0 b_N$, $a_0 = 1/\Upsilon_N$, and $\bar{\alpha}_{N,N} = 0$, gives $\mathbf{x}_N = (1/\Upsilon_N)\mathbf{x}_0 - W$, so the distance terms are

$$qB_{0,\star} - q\Upsilon_N^2 B_{N,\star} = q\,\mathbf{x}_0 \odot \mathbf{x}_0 - q\Upsilon_N^2\big(\tfrac{1}{\Upsilon_N}\mathbf{x}_0 - W\big) \odot \big(\tfrac{1}{\Upsilon_N}\mathbf{x}_0 - W\big) = 2q\Upsilon_N\, W \odot \mathbf{x}_0 - q\Upsilon_N^2\, W \odot W.$$

For the affine matrices $A_{i-1,i} = \mathbf{g}_i \odot (\mathbf{x}_{i-1} - \mathbf{x}_i)$ and $A_{\star,i} = -\mathbf{g}_i \odot \mathbf{x}_i$ we have $a_i(\mathbf{x}_{i-1} - \mathbf{x}_i) - \Delta a_i \mathbf{x}_i = a_i\mathbf{x}_{i-1} - a_{i+1}\mathbf{x}_i$, so (24) and (6) give

$$\sum_{i=1}^{N} a_i A_{i-1,i} + \sum_{i=1}^{N} \Delta a_i A_{\star,i} = \frac{1}{b_N}\sum_{i=1}^{N}(b_i - b_{i+1})\mathbf{g}_i \odot \mathbf{x}_0 + \sum_{i=1}^{N}\sum_{j=0}^{i-1}\big(a_{i+1}\alpha_{i,j} - a_i\alpha_{i-1,j}\big)\mathbf{g}_i \odot \mathbf{g}_j.$$

Note that the sum over $A_{\star,i}$ above starts at $i = 1$, whereas the corresponding multiplier sum in (25) starts at $i = 0$: the affine term $\Delta a_0 A_{\star,0} = -\Delta a_0\, \mathbf{g}_0 \odot \mathbf{x}_0$ has no partner $A_{i-1,i}$ in the first sum and, being a cross term with $\mathbf{x}_0$, is restored below when the $\mathbf{x}_0$ coefficients are collected. The finite difference (26) reads $a_{i+1}\bar{\alpha}_{i,j} - a_i\bar{\alpha}_{i-1,j} = c_\alpha\, \Delta\widehat{r}_i \cdot \Delta\widehat{r}_j$; since $\bar{\alpha}_{i-1,i-1} = 1/(1-q)$ while the inner sum uses $\alpha_{i-1,i-1} = 0$, that sum equals $c_\alpha \sum_{j=0}^{i-1}(\Delta\widehat{r}_i \cdot \Delta\widehat{r}_j)\, \mathbf{g}_i \odot \mathbf{g}_j + (a_i/(1-q))\mathbf{g}_i \odot \mathbf{g}_{i-1}$.

Multiplying the affine combination by $2\Upsilon_N$, restoring the boundary term $2\Upsilon_N \Delta a_0 A_{\star,0} = -2\Upsilon_N \Delta a_0 \mathbf{g}_0 \odot \mathbf{x}_0$, and expanding $C_{i-1,i} = \mathbf{g}_{i-1} \odot \mathbf{g}_{i-1} - 2\mathbf{g}_{i-1} \odot \mathbf{g}_i + \mathbf{g}_i \odot \mathbf{g}_i$ and $C_{\star,i} = \mathbf{g}_i \odot \mathbf{g}_i$, the term $(2\Upsilon_N/(1-q))\sum_i a_i \mathbf{g}_i \odot \mathbf{g}_{i-1}$ cancels the cocoercivity off-diagonal $-2\mathbf{g}_{i-1} \odot \mathbf{g}_i$, and the diagonal collects by index:

$$\begin{aligned}
\mathrm{LHS} = {} & 2\Upsilon_N\Big(qW + \tfrac{1}{b_N}\textstyle\sum_{i=1}^{N}(b_i - b_{i+1})\mathbf{g}_i - \Delta a_0 \mathbf{g}_0\Big) \odot \mathbf{x}_0 - q\Upsilon_N^2\, W \odot W \\
& + 2\Upsilon_N c_\alpha \sum_{i=1}^{N}\sum_{j=0}^{i-1}(\Delta\widehat{r}_i \cdot \Delta\widehat{r}_j)\, \mathbf{g}_i \odot \mathbf{g}_j \\
& + \frac{\Upsilon_N}{1-q}\Bigg[(2a_1 - a_0)\,\mathbf{g}_0 \odot \mathbf{g}_0 + \sum_{i=1}^{N-1}\underbrace{(a_{i+1} + a_i + \Delta a_i)}_{=\,2a_{i+1}}\,\mathbf{g}_i \odot \mathbf{g}_i + a_{N+1}\,\mathbf{g}_N \odot \mathbf{g}_N\Bigg].
\end{aligned}$$

The $\mathbf{x}_0$ coefficient vanishes: (27) gives $qW = (1/b_N)\sum_{i=0}^{N}\Delta b_i \mathbf{g}_i$, hence $qW + (1/b_N)\sum_{i=1}^{N}(b_i - b_{i+1})\mathbf{g}_i = (\Delta b_0/b_N)\mathbf{g}_0 = \Delta a_0 \mathbf{g}_0$, using $b_1 - b_0 = \Delta a_0 b_N$ from (6). The edge lengths (22) turn the diagonal bracket into $(\Upsilon_N/(qR_N^2))\sum_{i=0}^{N}\|\Delta\widehat{r}_i\|^2\, \mathbf{g}_i \odot \mathbf{g}_i$, so with $2\Upsilon_N c_\alpha = 2\Upsilon_N^2/(qb_N^2)$ and $b_N^2 = (1-q)R_N^2$ the diagonal and the double sum merge into $(\Upsilon_N^2/(qb_N^2))\sum_{i,j=0}^{N}(\Delta\widehat{r}_i \cdot \Delta\widehat{r}_j)\, \mathbf{g}_i \odot \mathbf{g}_j$. Finally $\Delta\widehat{r}_i \cdot \Delta\widehat{r}_j = \Delta a_i \Delta a_j + \Delta b_i \Delta b_j$, and the Δb part cancels $-q\Upsilon_N^2 W \odot W = -(\Upsilon_N^2/(qb_N^2))\sum_{i,j}\Delta b_i \Delta b_j\, \mathbf{g}_i \odot \mathbf{g}_j$, leaving

$$\mathrm{LHS} = \frac{\Upsilon_N^2}{qb_N^2}\sum_{i,j=0}^{N}\Delta a_i \Delta a_j\, \mathbf{g}_i \odot \mathbf{g}_j = \frac{\Upsilon_N^2}{qb_N^2}\Big(\sum_{i=0}^{N}\Delta a_i \mathbf{g}_i\Big) \odot \Big(\sum_{j=0}^{N}\Delta a_j \mathbf{g}_j\Big) = \frac{\Upsilon_N^2}{q(1-q)(\Upsilon_N^2 - 1)}\,\mathbf{z}\mathbf{z}^\top,$$

using $b_N^2 = (1-q)(\Upsilon_N^2 - 1)$ and $\mathbf{z} = (0, \Delta a_0, \ldots, \Delta a_N)^\top$. This proves (25); dividing by $2\Upsilon_N^2$ gives $Z = \frac{1}{2q(1-q)(\Upsilon_N^2 - 1)}\mathbf{z}\mathbf{z}^\top \succeq 0$. $\square$

This establishes all three conditions of (17). By weak duality, $f(x_N) - f_\star \leq \nu\,(f(x_0) - f_\star)$ with $\nu = 1/\Upsilon_N^2$, which is the bound (2). More precisely, this is the first inequality of (2) at the normalization $L = 1$ under which the certificate was assembled. For a general $f \in \mathcal{F}_{\mu,L}$ the rescaled function f/L belongs to $\mathcal{F}_{q,1}$, and recursion (31) applied to f generates the same iterates as when applied to f/L with $L = 1$, since $\nabla(f/L) = \frac{1}{L}\nabla f$ and the sequences $\{v_k\}$, $\{w_k\}$ depend only on q and N; the function-value ratio $(f(x_N) - f_\star)/(f(x_0) - f_\star)$ is invariant under this rescaling, so the first inequality of (2) holds for every $f \in \mathcal{F}_{\mu,L}$. The second inequality of (2) is Lemma 7.

It remains to show that the method certified above, namely the FSFOM whose transformed stepsizes are (19), is ITEM-f itself. This is the subject of the next section.

8 Equivalence of the method to (ITEM-f)

In this section we prove that the iterates generated by the method (11) with transformed stepsizes (19) matches the iterates of (ITEM-f). We first prove the equivalence to a FSFOM in the form (9). Then, we prove this FSFOM corresponds to (ITEM-f).

Three representations of an FSFOM. We first state and prove a general equivalence of three representations of a fixed-step first-order method: the h-matrix form, the memory-efficient form, and the momentum form.

Lemma 13 (Equivalence of representations)**.** *Let $v, w \in \mathbb{R}^N$ be vectors with $v_k \neq 0$. Let M be a fixed step first-order method with stepsize matrix $h_{i,j} = \delta_{i,j+1} + v_i w_j$ for $0 \leq j < i \leq N$, defined by*

$$x_{k+1} = x_k - \sum_{j=0}^{k} h_{k+1,j} \frac{1}{L} \nabla f(x_j), \quad 0 \leq k \leq N-1.$$

Then, M has a memory-efficient representation

$$\begin{aligned} z_{k+1} &= z_k - w_k \frac{1}{L}\nabla f(x_k), \\ x_{k+1} &= x_k^+ + v_{k+1} z_{k+1}, \end{aligned} \qquad 0 \leq k \leq N-1,$$

where $x_k^+ = x_k - (1/L)\nabla f(x_k)$ and $z_0 = 0$. Also, M has a momentum form

$$x_{k+1} = \begin{cases} x_0^+ + \eta_1\,(x_0^+ - x_0), & k = 0, \\ x_k^+ + \zeta_{k+1}\,(x_k^+ - x_{k-1}^+) + \eta_{k+1}\,(x_k^+ - x_k), & 1 \leq k \leq N-1, \end{cases}$$

with coefficients $\eta_1 = v_1 w_0$ and, for $1 \leq k \leq N-1$, $\zeta_{k+1} = v_{k+1}/v_k$ and $\eta_{k+1} = v_{k+1}\big(w_k - 1/v_k\big)$. In particular, the memory-efficient representation and the momentum form are equivalent to each other: started from the same x_0 (with $z_0 = 0$), they generate the same iterates $x_0, \ldots, x_N$.

Proof. Denote $g_k \triangleq (1/L)\nabla f(x_k)$, so that $x_k^+ = x_k - g_k$.

Memory-efficient form. Set $z_{k+1} \triangleq -\sum_{j=0}^{k} w_j g_j$, which starts at $z_0 = 0$ (empty sum) and satisfies $z_{k+1} = z_k - w_k g_k$. Since $\delta_{k+1,j+1} = \delta_{k,j}$, splitting the update of M along $h_{k+1,j} = \delta_{k,j} + v_{k+1} w_j$ gives

$$x_{k+1} = x_k - \sum_{j=0}^{k} \big(\delta_{k,j} + v_{k+1} w_j\big) g_j = x_k^+ + v_{k+1} z_{k+1},$$

which is the memory-efficient recursion.

Momentum form. Fix $1 \le k \le N-1$. The memory recursion at the previous step reads $x_k = x_{k-1}^+ + v_k z_k$, so $z_k = (x_k - x_{k-1}^+)/v_k$ (using $v_k \neq 0$). Substituting this and $g_k = x_k - x_k^+$ into $z_{k+1} = z_k - w_k g_k$ and then into $x_{k+1} = x_k^+ + v_{k+1} z_{k+1}$ gives

$$x_{k+1} = x_k^+ + v_{k+1}\left(\frac{x_k - x_{k-1}^+}{v_k} - w_k\big(x_k - x_k^+\big)\right);$$

regrouping via $x_k - x_{k-1}^+ = \big(x_k^+ - x_{k-1}^+\big) + \big(x_k - x_k^+\big)$ and collecting coefficients, a routine calculation, yields the momentum form with $\zeta_{k+1} = v_{k+1}/v_k$ and $\eta_{k+1} = v_{k+1}\big(w_k - 1/v_k\big)$. The first step is separate: $z_1 = -w_0 g_0$ gives $x_1 = x_0^+ - v_1 w_0 g_0 = x_0^+ + \eta_1\big(x_0^+ - x_0\big)$ with $\eta_1 = v_1 w_0$. $\square$

Equivalence of the method to a FSFOM. Recall the quantity s_k of (7) and define

$$\begin{aligned} v_k &\triangleq \frac{\phi_{N-k}}{(1-q)^k a_k}, && 0 \le k \le N, \\ w_k &\triangleq \frac{(1-q)^k a_{k+1}\phi_k}{\Upsilon_N}, && 0 \le k \le N-1. \end{aligned} \tag{28}$$

We define the stepsizes as

$$h_{i,j} \triangleq \delta_{i,j+1} + v_i w_j, \quad 0 \le j < i \le N. \tag{29}$$

Since a_k and ϕ_{N-k} are positive, $v_k \neq 0$.

To show the equivalence of the method to the FSFOM in the form (9) with these stepsizes (29), we prove that the reparametrization (13) of the transformed stepsizes α of (19) produces $h_{i,j} = \delta_{i,j+1} + v_i w_j$.

We organize the proof around two planar vectors read off the construction, the *tangent vector* $\widehat{t}_k$ and the *memory vector* $\widehat{m}_k$:

$$\begin{aligned} \widehat{t}_k &= \begin{pmatrix} \Upsilon_N b_k \\ 1 - \Upsilon_N a_k \end{pmatrix}, && 0 \le k \le N, \\ \widehat{m}_k &= \frac{\sqrt{q}(1-q)^k}{\Upsilon_N}\,\widehat{t}_k, && 1 \le k \le N. \end{aligned} \tag{30}$$

Lemma 14. *For the vectors in* (18) *and* (30)*, the following identities hold:*

(1) (Primary step.) *For* $1 \le k \le N$, $\widehat{u}_{k+1} = (1-q)\widehat{u}_k + v_k\,\widehat{m}_k$.

(2) *For* $0 \le k \le N-1$, $c_\alpha \widehat{m}_{k+1} \cdot \Delta\widehat{r}_k = w_k$.

(3) (Auxiliary step.) *For* $1 \le k \le N-1$, $\widehat{m}_{k+1} = \widehat{m}_k - q\,w_k\,\widehat{u}_{k+1}$.

Proof. (1). Let $\widehat{n}_k \triangleq \Upsilon_N \widehat{r}_k + (R_N^2, 0)^\top = (\Upsilon_N a_k - 1,\ \Upsilon_N b_k)^\top$ be the normal, so that the tangent $\widehat{t}_k$ is its $90°$ rotation and $\widehat{n}_k \cdot \widehat{t}_k = 0$. Since $\|\widehat{r}_k\|^2 = R_N^2$,

$$\widehat{n}_k \cdot \widehat{u}_k = \frac{\Upsilon_N \|\widehat{r}_k\|^2 + R_N^2(a_k - \Upsilon_N)}{a_k} = R_N^2,$$

and by the second line of (1),

$$\widehat{n}_k \cdot \widehat{u}_{k+1} = \frac{\Upsilon_N\,\widehat{r}_k \cdot \widehat{r}_{k+1} + R_N^2(a_{k+1} - \Upsilon_N)}{a_{k+1}} = (1-q)R_N^2.$$

Hence $\widehat{n}_k \cdot \big(\widehat{u}_{k+1} - (1-q)\widehat{u}_k\big) = 0$, so $\widehat{u}_{k+1} - (1-q)\widehat{u}_k = \mu_k \widehat{t}_k$ for a scalar μ_k. Taking $\det(\cdot, \widehat{u}_k)$ and using $\det(\widehat{t}_k, \widehat{u}_k) = \widehat{n}_k \cdot \widehat{u}_k = R_N^2$, Lemma 8 gives, for every $1 \le k \le N$,

$$\begin{aligned}\mu_k R_N^2 = \det(\widehat{u}_{k+1}, \widehat{u}_k) &= \frac{\det(\widehat{r}_{k+1}, \widehat{r}_k)}{a_k a_{k+1}} \\ &= \frac{R_N^2 s_k}{a_k a_{k+1}} = \frac{\sqrt{q}\, R_N^2 \phi_{N-k}}{\Upsilon_N a_k}.\end{aligned}$$

Thus $\mu_k = \sqrt{q}\,\phi_{N-k}/(\Upsilon_N a_k)$, and the definitions of v_k and $\widehat{m}_k$ give $\mu_k \widehat{t}_k = v_k \widehat{m}_k$.

(2). Direct expansion and the coordinate symmetry give

$$\begin{aligned}\widehat{t}_k \cdot \Delta\widehat{r}_k &= (1 - \Upsilon_N a_k) b_{k+1} - b_k (1 - \Upsilon_N a_{k+1}) \\ &= \det(\widehat{u}_{N-k+1}, \widehat{u}_{N-k}) \\ &= \frac{R_N^2 s_{N-k}}{a_{N-k} a_{N-k+1}} = \frac{\sqrt{q}\, R_N^2 a_{k+1} \phi_k}{\Upsilon_N},\end{aligned}$$

where the third equality is Lemma 8 at index $N-k$ (which lies in $\{1, \dots, N\}$ for $0 \le k \le N-1$), and the second and last use the symmetry (6). Moreover, $\widehat{t}_{k+1} - \widehat{t}_k = (\Upsilon_N \Delta b_k, -\Upsilon_N \Delta a_k)^\top$ is orthogonal to $\Delta\widehat{r}_k = (\Delta a_k, \Delta b_k)^\top$, so $\widehat{t}_{k+1} \cdot \Delta\widehat{r}_k = \widehat{t}_k \cdot \Delta\widehat{r}_k$. Substituting the definitions of c_α, $\widehat{m}_{k+1}$, and w_k now gives $c_\alpha \widehat{m}_{k+1} \cdot \Delta\widehat{r}_k = w_k$.

(3). The calculation in (1) gives

$$\det(\widehat{m}_k, \widehat{u}_k) = \frac{\sqrt{q}(1-q)^k R_N^2}{\Upsilon_N}, \qquad 1 \le k \le N.$$

Consequently, using (1),

$$\begin{aligned}&\det(\widehat{m}_{k+1} - \widehat{m}_k, \widehat{u}_{k+1}) \\ &\quad = \det(\widehat{m}_{k+1}, \widehat{u}_{k+1}) - (1-q)\det(\widehat{m}_k, \widehat{u}_k) = 0,\end{aligned}$$

so $\widehat{m}_{k+1} - \widehat{m}_k = \sigma_k \widehat{u}_{k+1}$ for a scalar σ_k. Dotting with $\Delta\widehat{r}_k$, the definitions of $\widehat{m}_k$ and $\widehat{m}_{k+1}$, together with $\widehat{t}_{k+1} \cdot \Delta\widehat{r}_k = \widehat{t}_k \cdot \Delta\widehat{r}_k$, show that (2) also gives

$$c_\alpha \widehat{m}_k \cdot \Delta\widehat{r}_k = \frac{w_k}{1-q}.$$

Together with (2) and $c_\alpha \widehat{u}_{k+1} \cdot \Delta\widehat{r}_k = 1/(1-q)$ from (23), this yields

$$\frac{\sigma_k}{1-q} = w_k - \frac{w_k}{1-q} = -\frac{q w_k}{1-q}.$$

Hence $\sigma_k = -q w_k$. □

By Lemma 13, the FSFOM with stepsizes (29) is equivalent to the following memory-efficient representation. Starting from $z_0 = 0$, for $0 \le k \le N-1$, the equivalent update is:

$$\begin{aligned} z_{k+1} &= z_k - w_k \frac{1}{L} \nabla f(x_k), \\ x_{k+1} &= x_k^+ + v_{k+1} z_{k+1}. \end{aligned} \tag{31}$$

We briefly note that the steps (1) and (3) in Lemma 14 represent the input-free form of (31): with vanishing transformed gradients $\tilde{g}_k = 0$ so that $\nabla f(x_k) = q x_k$, the pair $(x_k, z_k) = (\widehat{u}_{k+1}, \widehat{m}_k)$ satisfies (31) for $1 \le k \le N-1$.

We now prove that the reparametrization (13) of the transformed stepsizes of (19) produces the stepsizes (29). We state the definitions once more for readability.

Lemma 15. *Let* $\{\alpha_{i,j}\}_{0\le j<i\le N}$ *be the transformed stepsizes of* (19),

$$\alpha_{i,j} = c_\alpha\, \widehat{u}_{i+1}\cdot \Delta\widehat{r}_j,$$
$$c_\alpha = \frac{\Upsilon_N}{q(1-q)R_N^2},$$

and let the FSFOM stepsizes $\{h_{i,j}\}_{0\le j<i\le N}$ *be recovered from them by the reparametrization* (13),

$$h_{i,j} = \begin{cases} \alpha_{i,i-1}, & j=i-1,\\ \alpha_{i,j}-\alpha_{i-1,j}+q\displaystyle\sum_{k=j+1}^{i-1} h_{i,k}\,\alpha_{k,j}, & 0\le j\le i-2. \end{cases}$$

Then, with the scalar sequences of (28)*, the recovered stepsizes have the closed form*

$$h_{i,j} = \delta_{i,j+1} + v_i w_j, \quad 0\le j<i\le N.$$

Proof. Diagonal. Since $\Upsilon_N/qR_N^2 = (1-q)c_\alpha$, (23) gives $c_\alpha(1-q)\,\widehat{u}_i\cdot\Delta\widehat{r}_{i-1} = 1$ for $1\le i\le N$. By Lemma 14, we have

$$\alpha_{i,i-1} - 1 = c_\alpha\big(\widehat{u}_{i+1} - (1-q)\widehat{u}_i\big)\cdot\Delta\widehat{r}_{i-1} = c_\alpha v_i \widehat{m}_i\cdot\Delta\widehat{r}_{i-1} = v_i w_{i-1}, \tag{32}$$

so the diagonal rule is $h_{i,i-1} = \alpha_{i,i-1} = 1 + v_i w_{i-1}$.

Off-diagonal. Fix $0\le j\le i-2$ (so $i\ge 2$). By Lemma 14, $\widehat{u}_{i+1} - (1-q)\widehat{u}_i = v_i\widehat{m}_i$ and $q\,w_k\widehat{u}_{k+1} = \widehat{m}_k - \widehat{m}_{k+1}$, so the sum telescopes:

$$\widehat{u}_{i+1} - (1-q)\widehat{u}_i + q\,v_i\sum_{k=j+1}^{i-1} w_k\,\widehat{u}_{k+1} = v_i\Big(\widehat{m}_i + \sum_{k=j+1}^{i-1}(\widehat{m}_k - \widehat{m}_{k+1})\Big) = v_i\,\widehat{m}_{j+1}.$$

Apply $c_\alpha\langle\,\cdot\,,\Delta\widehat{r}_j\rangle$ to this identity. On the left $c_\alpha\widehat{u}_{i+1}\cdot\Delta\widehat{r}_j = \alpha_{i,j}$ and $c_\alpha\widehat{u}_{k+1}\cdot\Delta\widehat{r}_j = \alpha_{k,j}$; on the right, by Lemma 14 and its second identity, $c_\alpha v_i\widehat{m}_{j+1}\cdot\Delta\widehat{r}_j = v_i w_j$. This gives

$$\alpha_{i,j} - (1-q)\alpha_{i-1,j} + q\,v_i\sum_{k=j+1}^{i-1} w_k\,\alpha_{k,j} = \delta_{i,j+1} + v_i w_j, \tag{33}$$

valid for all $0\le j<i\le N$: for $j=i-1$ the term $\alpha_{i-1,j}$ is absent and (32) supplies $\delta_{i,j+1}$. Set $\widetilde{h}_{i,j} = \delta_{i,j+1} + v_i w_j$. Because $q\sum_{k=j+1}^{i-1}\widetilde{h}_{i,k}\alpha_{k,j} = q\,\alpha_{i-1,j} + q\,v_i\sum_{k=j+1}^{i-1} w_k\alpha_{k,j}$ (the δ in $\widetilde{h}_{i,k}$ selecting $k=i-1$), identity (33) reads $\widetilde{h}_{i,j} = \alpha_{i,j} - \alpha_{i-1,j} + q\sum_{k=j+1}^{i-1}\widetilde{h}_{i,k}\alpha_{k,j}$, which is exactly the triangular recursion (13) defining h. As that recursion determines each row uniquely from right to left, $h_{i,j} = \widetilde{h}_{i,j}$. $\square$

Equivalence of the FSFOM to (ITEM-f)**.** We now prove the FSFOM with stepsizes (29) corresponds to (ITEM-f). Direct calculation from (28) gives

$$\frac{v_{k+1}}{v_k} = \frac{a_k\,\phi_{N-k-1}}{(1-q)\,a_{k+1}\,\phi_{N-k}}, \qquad 0\le k\le N-1,$$

which is the momentum coefficient of ITEM-f. For the correction coefficient, the factors $(1-q)^k$ cancel in the product

$$v_{k+1}w_k = \frac{\phi_{N-k-1}}{(1-q)^{k+1}a_{k+1}}\cdot\frac{(1-q)^k a_{k+1}\phi_k}{\Upsilon_N} = \frac{\phi_k\,\phi_{N-k-1}}{(1-q)\,\Upsilon_N},$$

and therefore

$$v_{k+1}w_k - \frac{v_{k+1}}{v_k} = \frac{\phi_{N-k-1}}{(1-q)\,\phi_{N-k}}\left(\frac{\phi_k\,\phi_{N-k}}{\Upsilon_N} - \frac{a_k}{a_{k+1}}\right) = \frac{\phi_{N-k-1}}{(1-q)\,\phi_{N-k}}\,(1-q) = \frac{\phi_{N-k-1}}{\phi_{N-k}}, \qquad 0 \le k \le N-1,$$

where the middle equality substitutes Lemma 9 (at $k=0$ through its stated extension). The two coefficient identities show that (ITEM-f) is the momentum form of Lemma 13 for $1 \le k \le N-1$. At $k=0$, the convention $x_{-1}^{+} = x_0$ makes the two coefficients add to $v_1 w_0$, which is the $k=0$ case of the momentum form.

Therefore the rate $\nu = 1/\Upsilon_N^2$ proved in Section 7 applies to the iterates of (ITEM-f), and with the relaxation in Lemma 7, this proves Theorem 1.

9 How the certificate was found

Remark 1. The certificate verified in this document was not constructed by hand. Following the branch-and-bound performance estimation programming (BnB-PEP) methodology [2], the transformed stepsizes are treated as decision variables alongside the inner-dual variables: substituting the dual (17) for the inner maximization turns the design of an optimal method into a single minimization problem, jointly over α and (λ, ν, Z), and replacing the semidefinite constraint $Z \succeq 0$ with the Cholesky factorization $Z = VV^{\intercal}$, where V is lower triangular with nonnegative diagonals [4, Corollary 7.2.9], makes the joint problem amenable to nonlinear and spatial branch-and-bound solvers. Solving this problem numerically for small N and fitting symbolic expressions to the resulting solutions, as described in the companion paper [1], produced the analytic formulas of Section 6, including the geometric construction of Lemma 1. The verification given in Sections 7 and 8 is entirely classical and independent of this discovery process.

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